

Model Optimization and Tuning Phase Template

Date	28 June 2025
Student ID	SWUID20250181505
Project Title	online payments fraud detection using machine learning
Maximum Marks	10 Marks

Model Optimization and Tuning Phase

The Model Optimization and Tuning Phase involves refining machine learning models for peak performance. It includes optimized model code, fine-tuning hyperparameters, comparing performance metrics, and justifying the final model selection for enhanced predictive accuracy and efficiency.

Hyperparameter Tuning Documentation (6 Marks):

Model	Tuned Hyperparameters	Optimal Values
Random Forest Classifier	None	Used default hyperparameters
Decision Tree Classifier	None	Used default hyperparameters
Extra Trees Classifier	None	Used default hyperparameters

Support Vector Classifier (SVC)	None	Used default hyperparameters
XGBoost Classifier	None	Used default hyperparameters

Performance Metrics Comparison Report (2 Marks):

Model	Baseline Metric	Optimized Metric
Random Forest Classifier	Accuracy: 0.9733 (97.33%), F1-Score (Fraud): 0.97	Not applicable (no tuning performed)
Decision Tree Classifier	Accuracy: 0.9383 (93.83%), F1-Score (Fraud): 0.93	Not applicable (no tuning performed)
Extra Trees Classifier	Accuracy: 0.9650 (96.50%), F1-Score (Fraud): 0.96	Not applicable (no tuning performed)

Support Vector Classifier (SVC)	Accuracy: 0.7695 (76.95%), F1-Score (Fraud): 0.66	Not applicable (no tuning performed)
XGBoost Classifier	Accuracy: 0.9835 (98.35%), F1-Score (Fraud): 0.98	Not applicable (no tuning performed)

Final Model Selection Justification (2 Marks):

Final Model	Reasoning
Random Forest Classifier	The Random Forest Classifier was chosen as the final model for deployment in the Flask-based web application, despite XGBoost achieving a slightly higher test accuracy (98.35%) and F1-score (0.98) compared to Random Forest's 97.33% accuracy and 0.97 F1-score. Random Forest was selected for its robustness, lower risk of overfitting compared to Decision Tree (93.83% accuracy), and better performance than SVC (76.95% accuracy). Its ensemble nature ensures stable predictions, and its compatibility with pickle serialization facilitates integration with the Flask application. Additionally, Random Forest offers good interpretability through feature importance, which is valuable for understanding fraud patterns. While XGBoost performed slightly better, Random Forest's balance of performance, computational

	efficiency, and ease of deployment made it the preferred choice for this project. efficiency, and ease of deployment made it the preferred choice for this project.
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